



tablets, and other devices. It's an open source operating system based on the Android Open Source Project code that can provide new features and security updates to older devices, among other things.

LineageOS 19, the first stable version of the operating system based on Android 12 and Android 12L, has now been released by the operating system's developers. The current version of LineageOS includes, among other things, a modified version of Android 12's Material You design language, as well as improved privacy and accessibility features. LineageOS 19 also includes Android security fixes released through April 2022 (and backported to LineageOS 16, 17, and 18).

Due to modifications in the Android Open Source Project code, some older Android devices will not support LineageOS 19. Basically, if your phone, tablet, or other device runs on Linux kernel 3.18 or older, you won't be seeing an official build of LineageOS 19 anytime soon.

Bloomberg open sources Memray

Memray is a memory profiler that was developed at Bloomberg and is now open source. It can track memory allocations in Python code, including native extensions and the Python interpreter itself. Memory profiling is a strong tool for understanding how a program utilises memory and, as a result, detecting memory leaks or determining which areas of the program consume the most memory.



In contrast to sampling memory profilers like py-spy, Memray can track every function call, including calls into C/C++ libraries, and display the call stack in detail. Bloomberg claims that this does not come at the sacrifice of performance, with profiling only slowing down interpreted code by a little amount. However, native code profiling is slower and must be enabled directly.

According to Yury Selivanov, co-founder and CEO of EdgeDB, the tool gives previously unavailable insights into Python applications.

Spotify moves to the rhythm of open source software

After two years as an open source project, Spotify, the music-streaming behemoth has open sourced a number of its projects, including Backstage, which was recently accepted as an incubating project at the Cloud Native Computing Foundation (CNCF). In addition, the corporation recently joined the Open Source Security Foundation, established an open source program office, and is now developing a fund to promote independent open source projects.

Cloudflare to create new API standards

Cloudflare, Inc., the Internet security, performance, and reliability company, is collaborating with Deno and individual core contributors of the Node.js open source project, which brings together three of the largest JavaScript environments, to give developers flexibility and choice while creating the standards for the future of edge computing. The project will seek to ensure that code produced in one environment will operate in another by working around a shared set of standards. Any developer will be able to write and run code that adheres to the set of standards—as well as simply transfer it—across Cloudflare Workers, Deno, and Node.js without having to redesign an application.

“The future of JavaScript is Universal, the ability to move JavaScript between many environments. That future needs to be intentionally designed,” said Myles Borins, member of the Node.js Technical Steering Committee. “Historically, developers have had to rely on tacit knowledge and build tools to share code across the various JavaScript runtimes. The promise of ‘write once, run anywhere’ will only be possible with collaborative initiatives like the Web-interoperable Runtimes Community Group.”

“Developers are asking Node.js for more interoperability with APIs defined by Web standards,” said Matteo Collina, chief software architect, NearForm and Node.js Technical Steering Committee member. “However, those APIs have not been designed with servers in mind and might require some adaptations. The WinterCG group’s goal is to define how those APIs would behave on servers, paving the way for better interoperability between runtimes.”